RELATE

Week 6

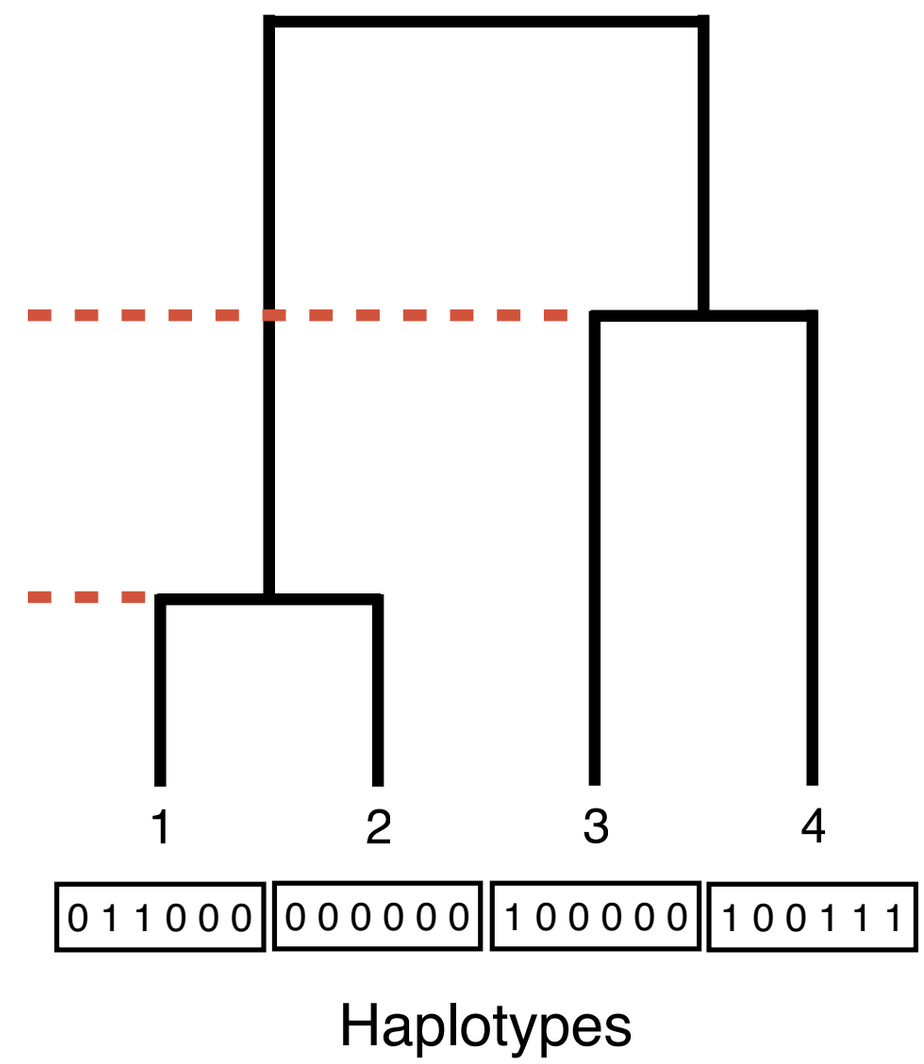
Kasper Munch

Sequence clustering

Of non-recombining sequences

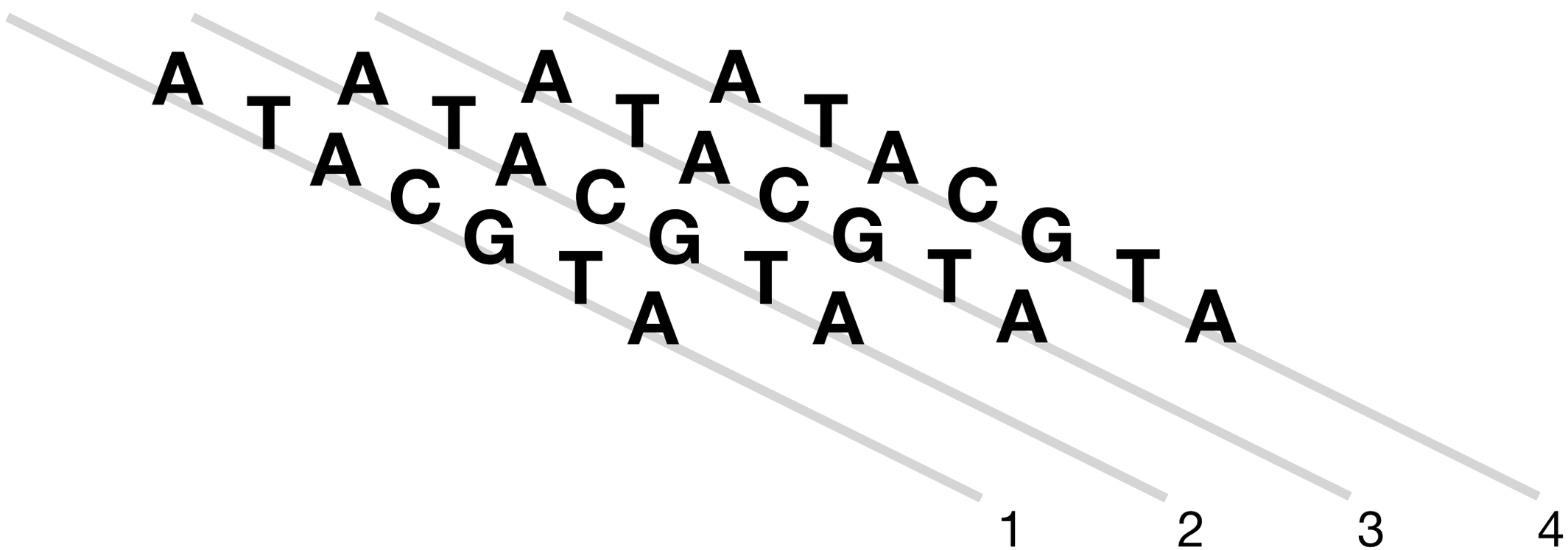
Distance matrix

	1	2	3	4
1	-	2	2	2
2	0	-	0	0
3	1	1	-	1
4	4	3	3	-

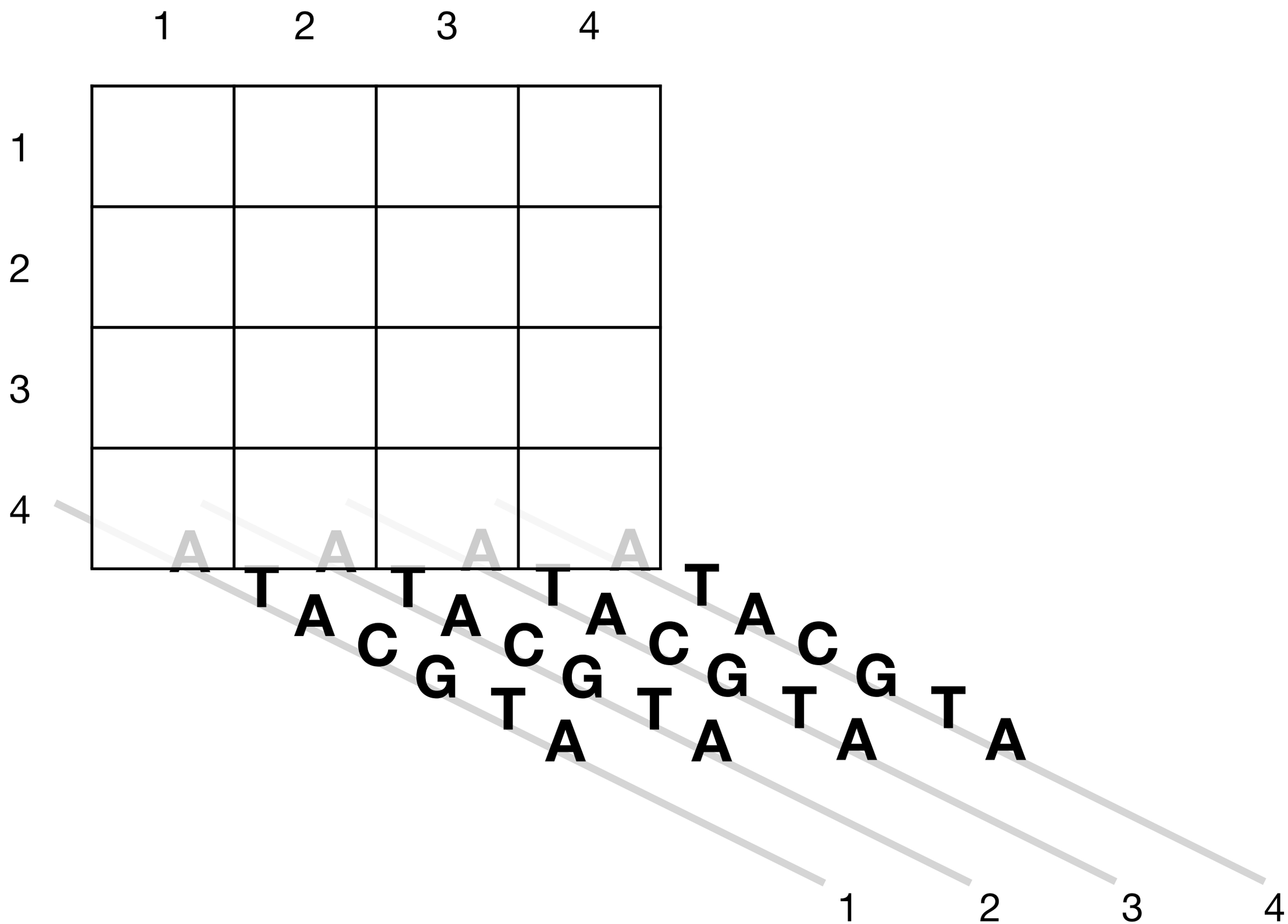


Sequence clustering

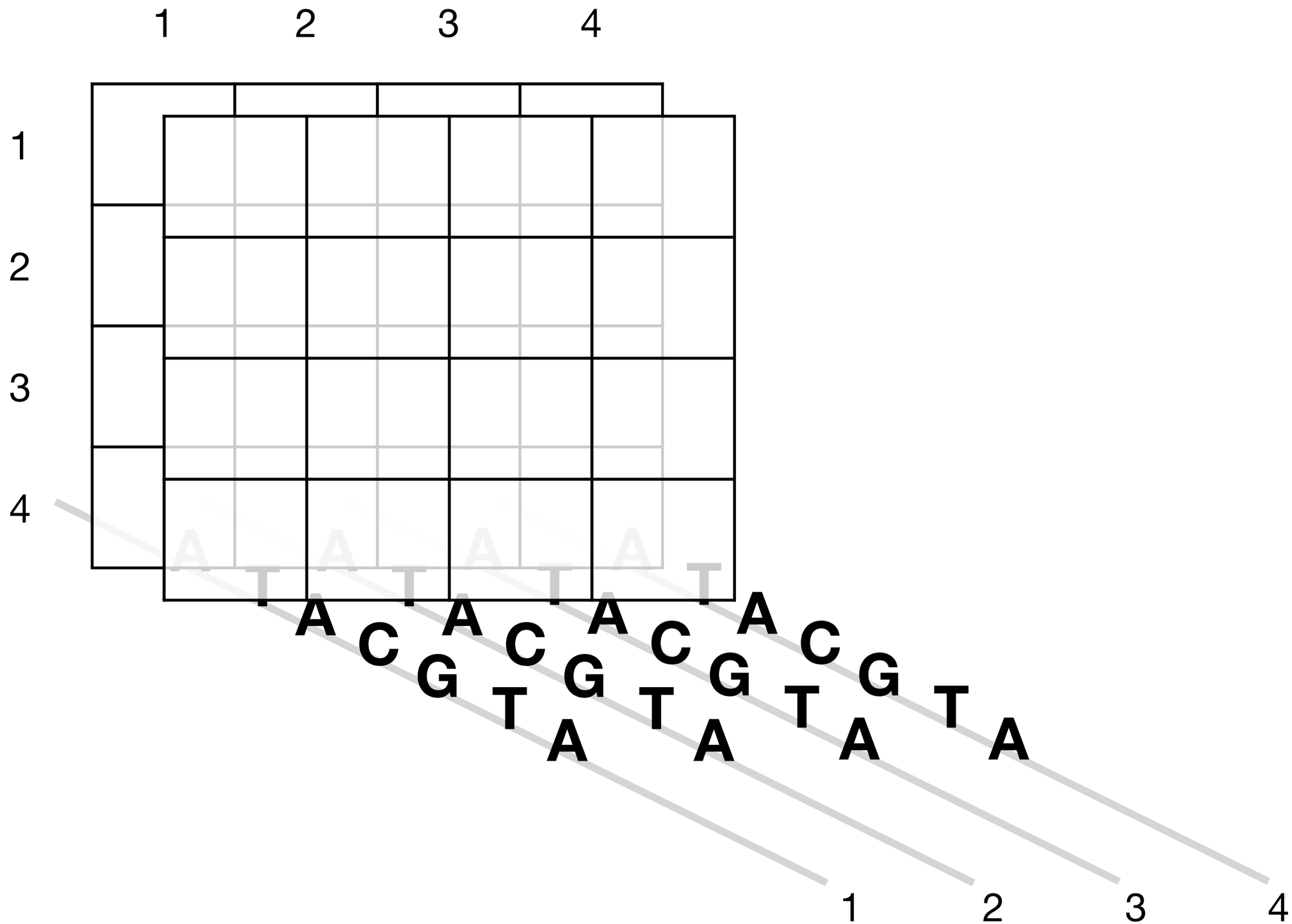
Of recombining sequences



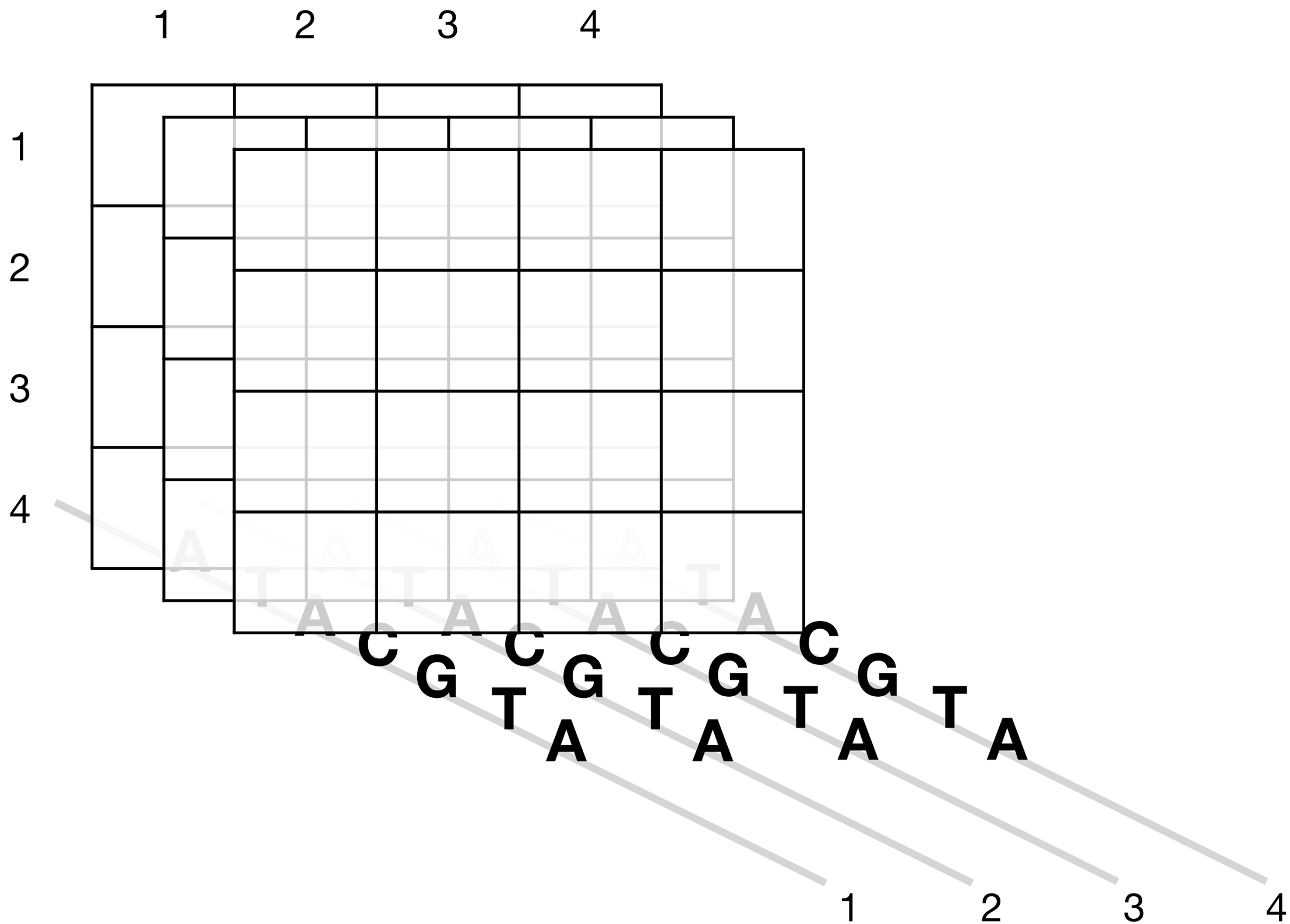
Sequence clustering



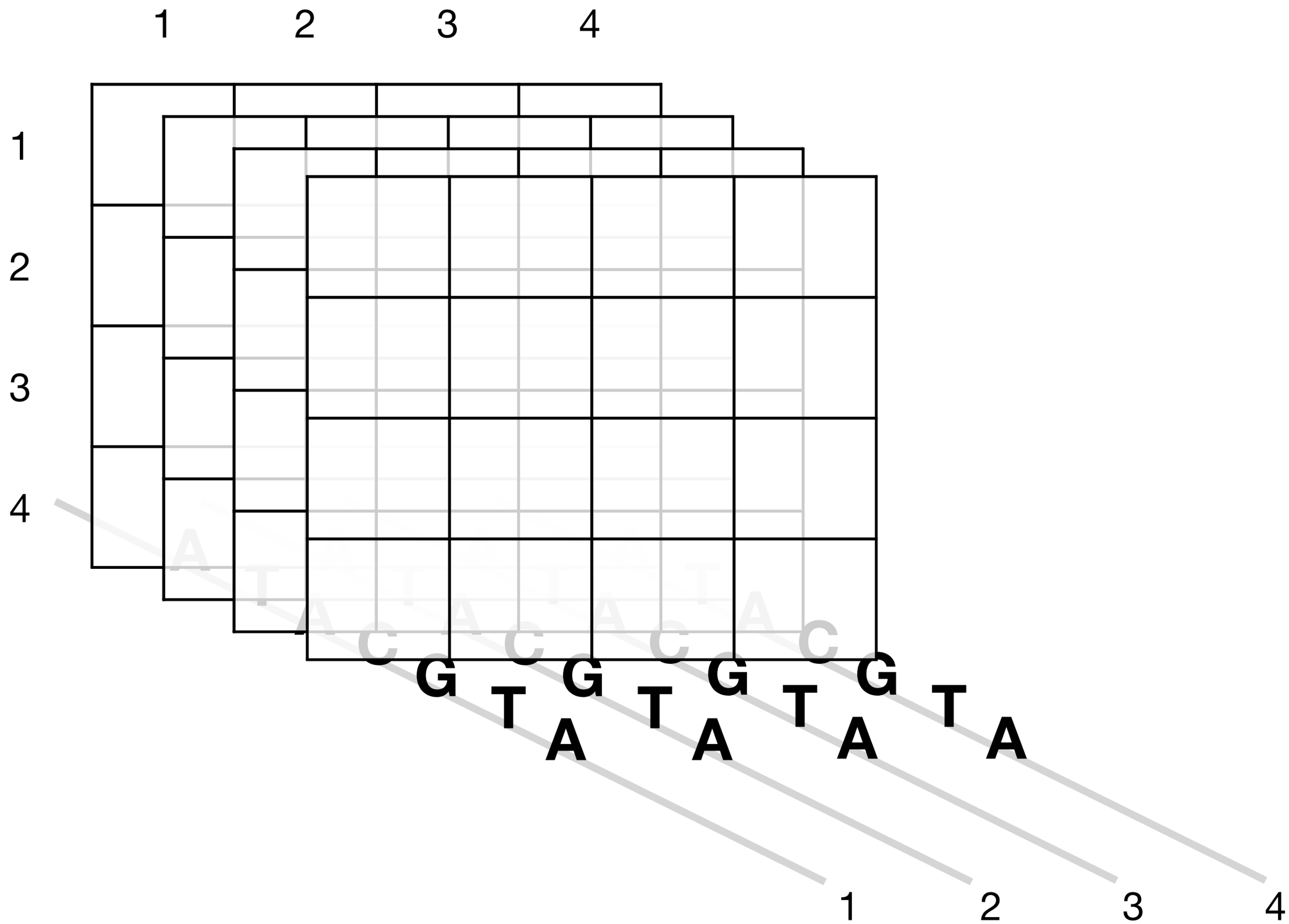
Sequence clustering



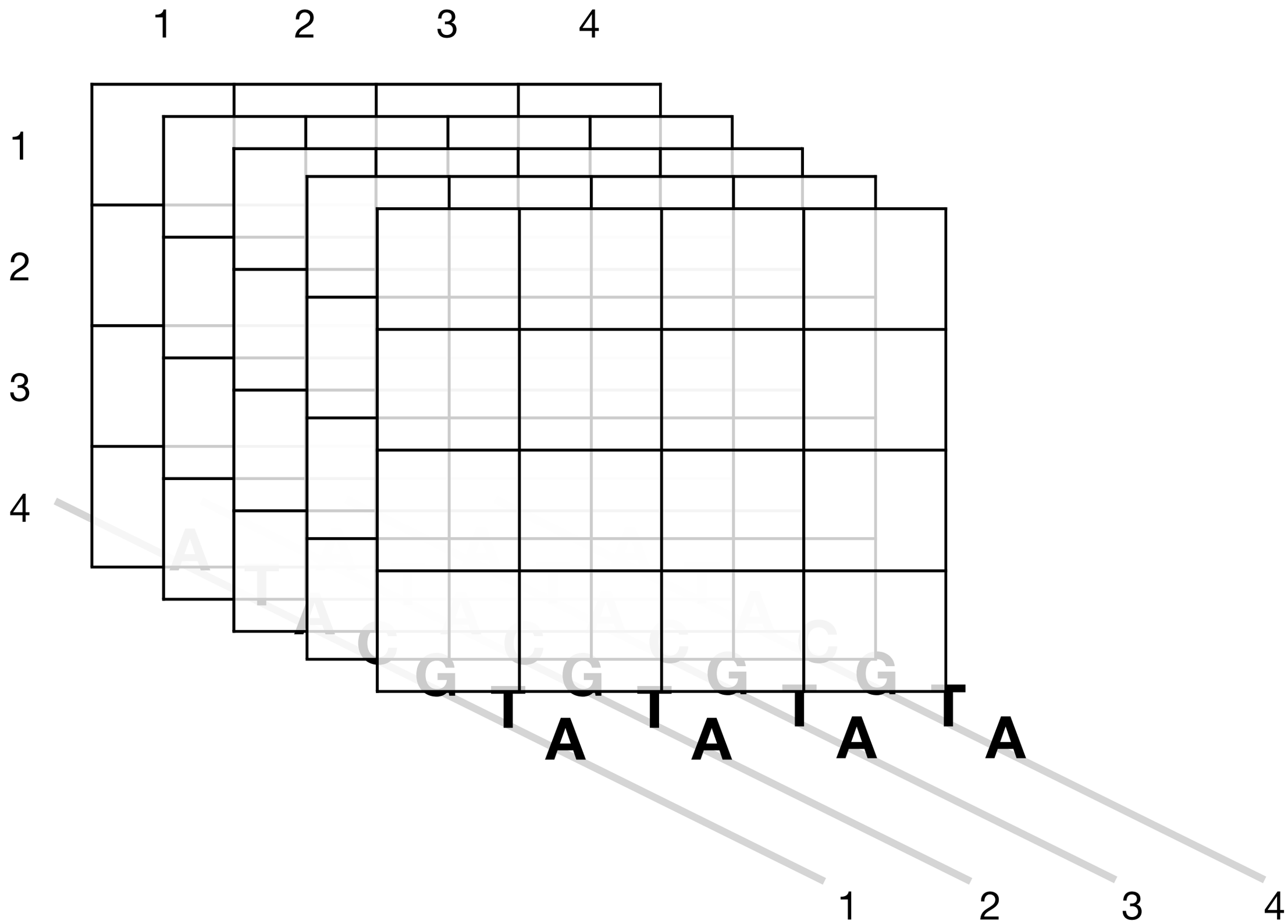
Sequence clustering



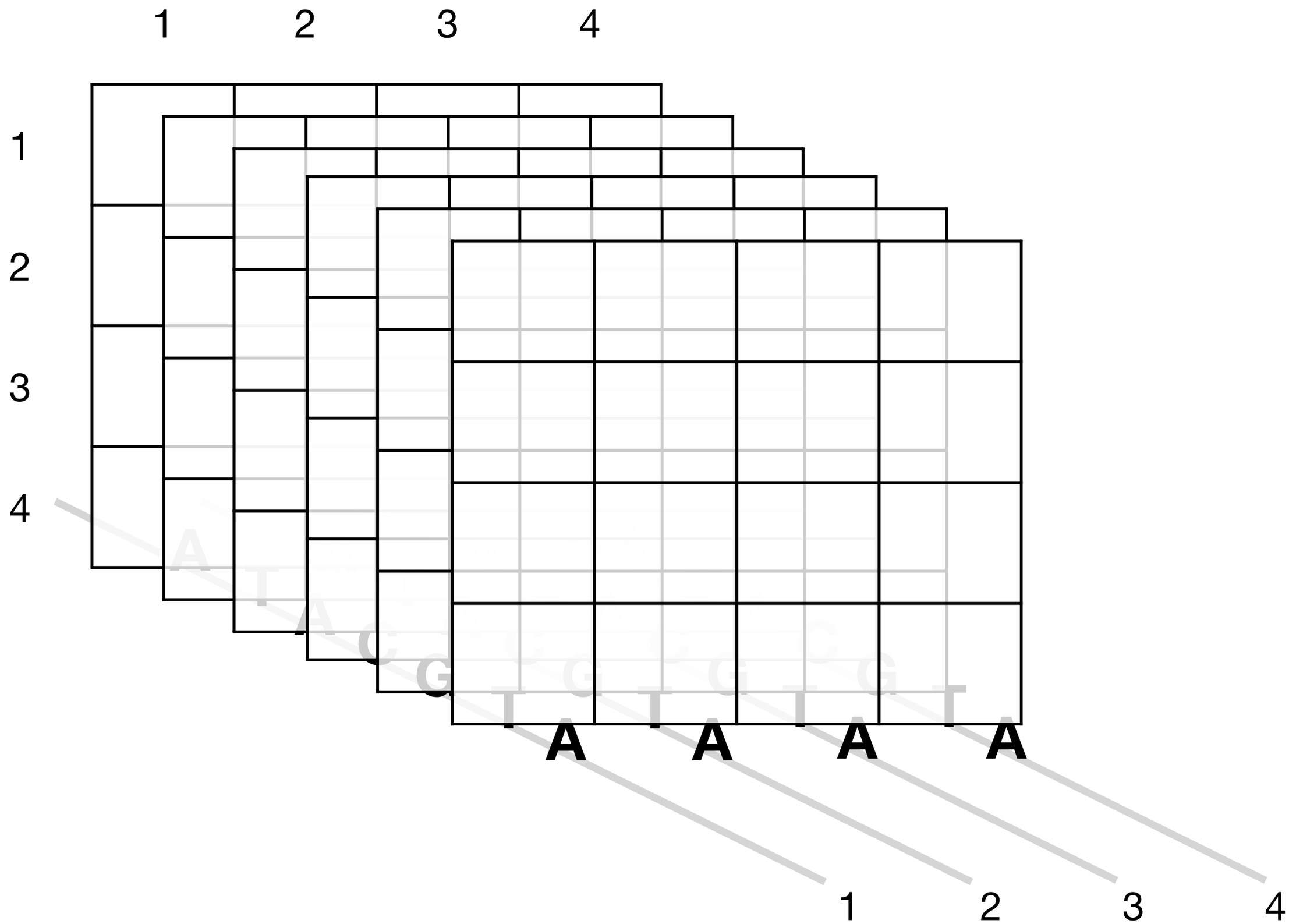
Sequence clustering



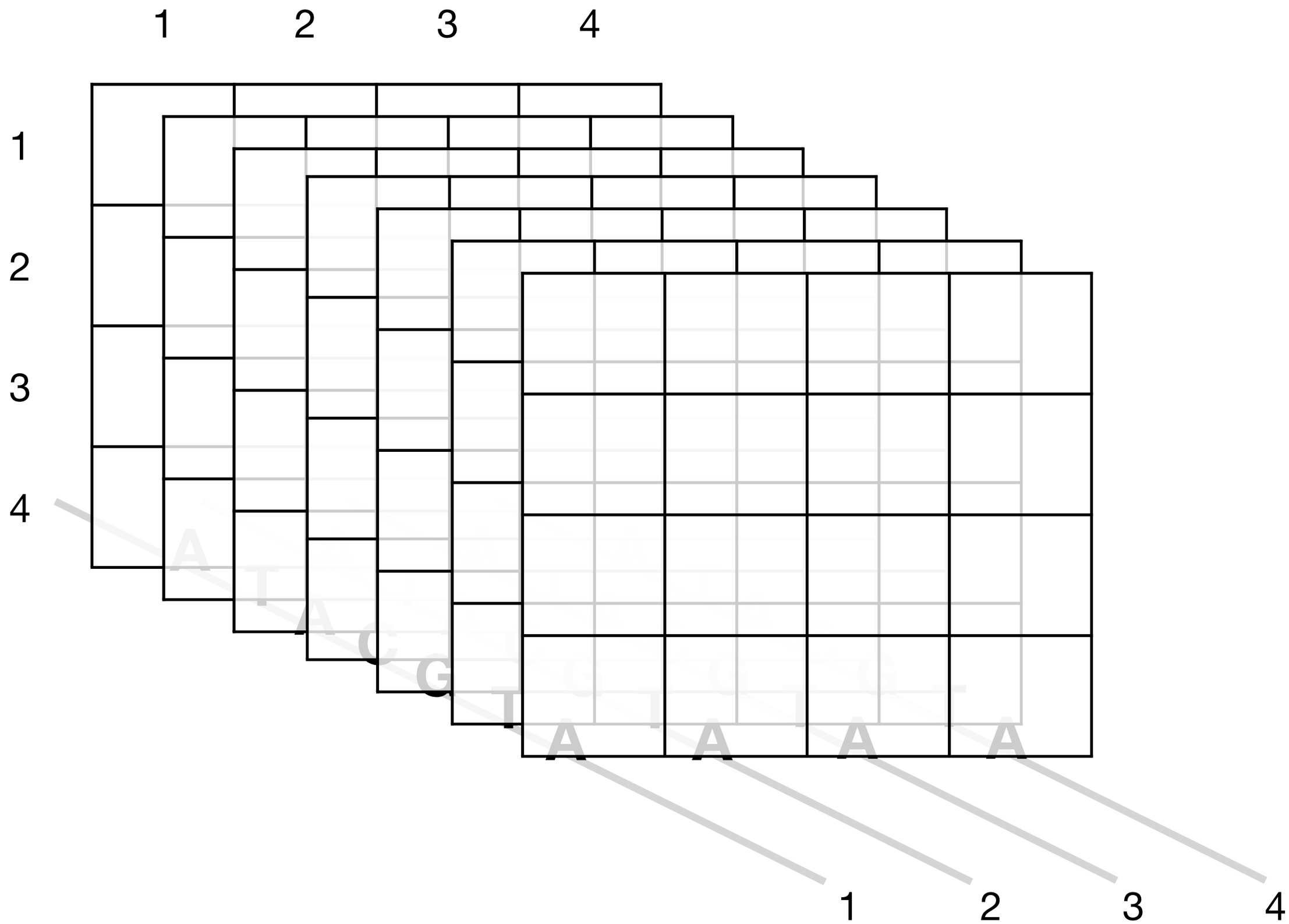
Sequence clustering



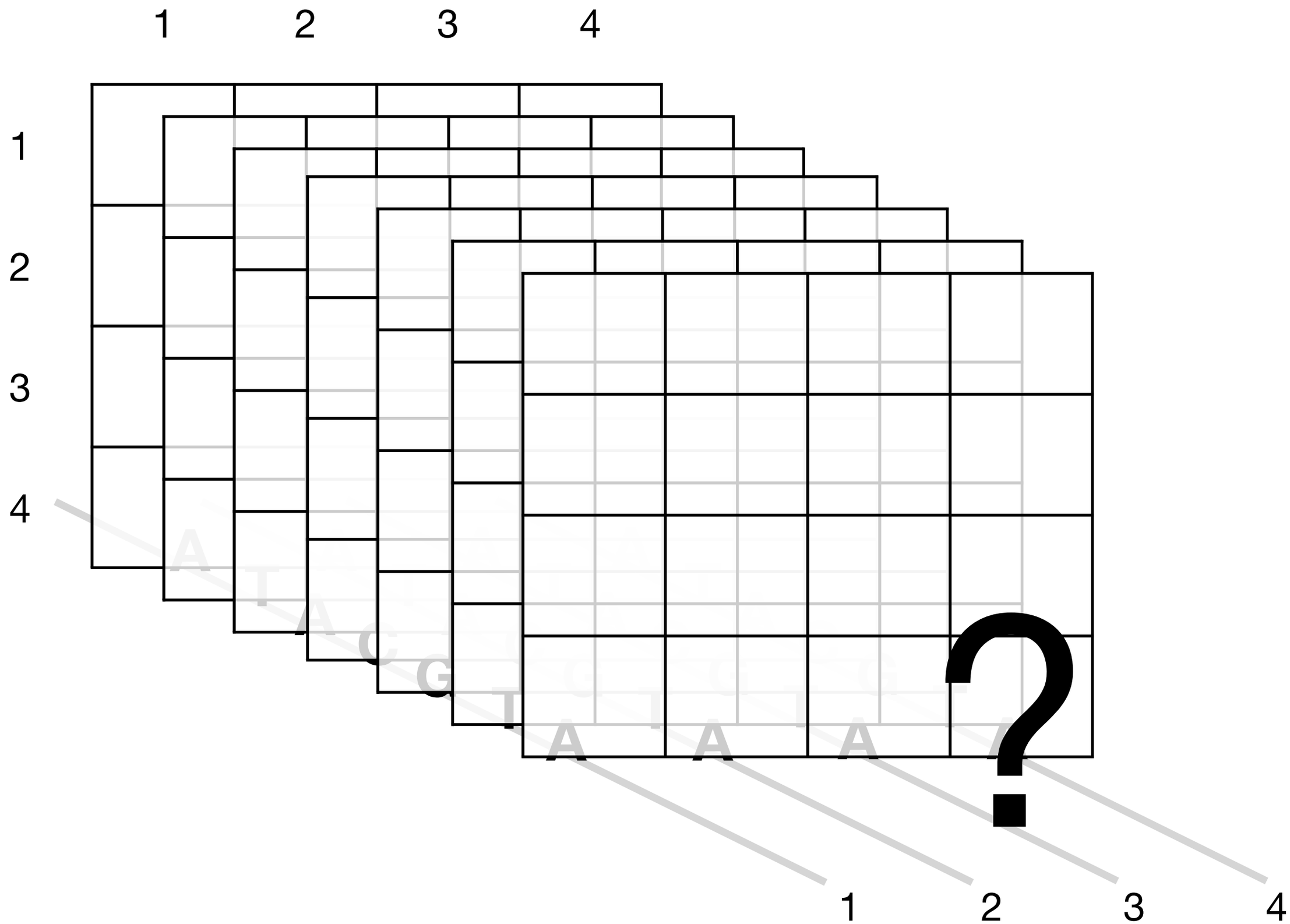
Sequence clustering



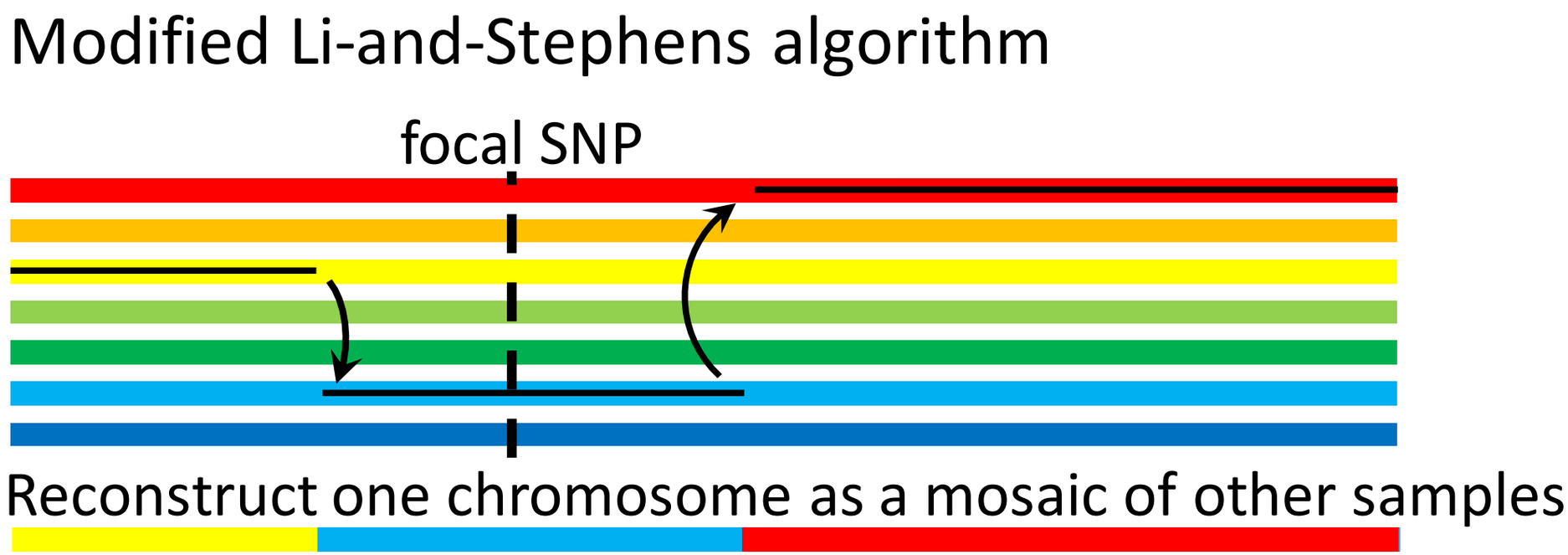
Sequence clustering



Sequence clustering

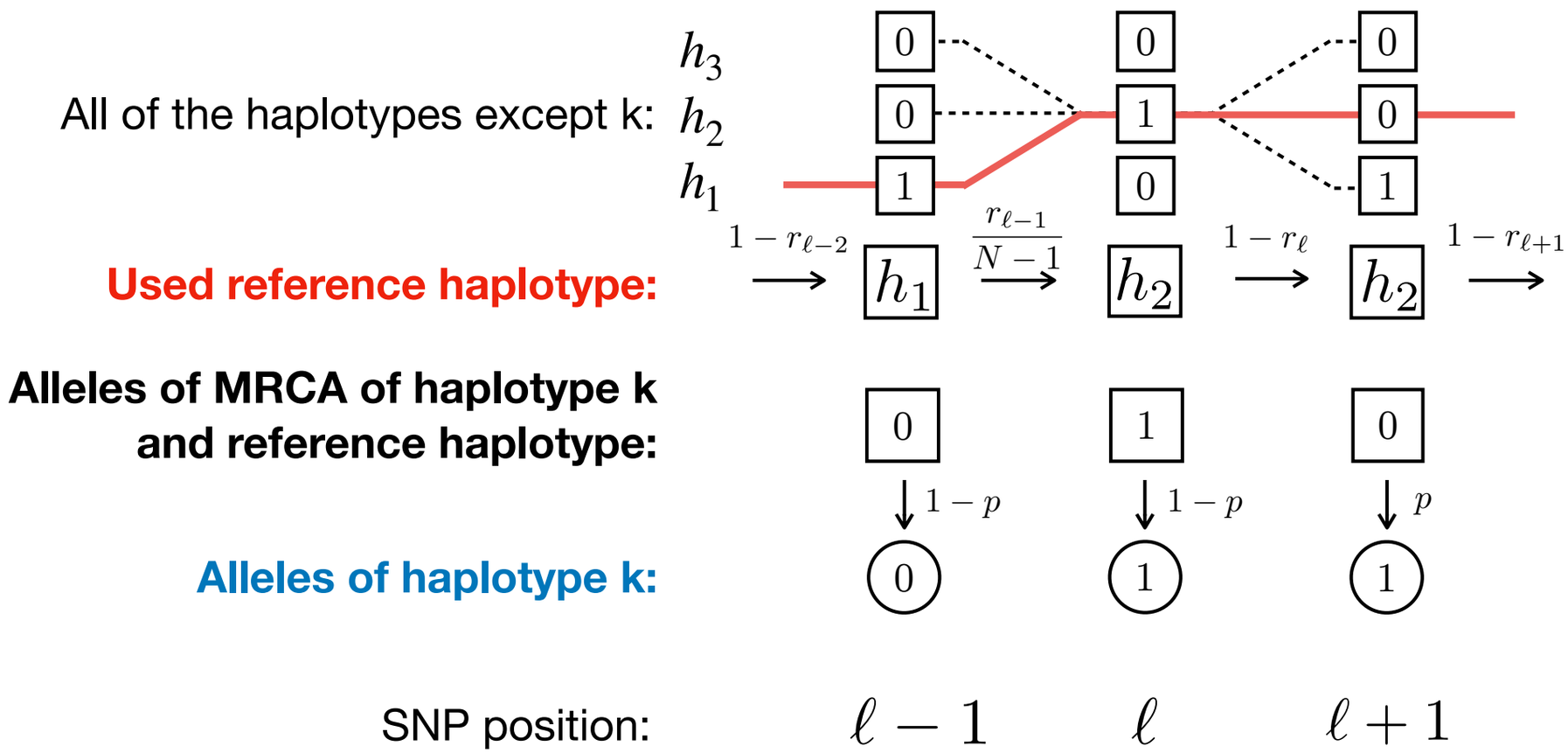


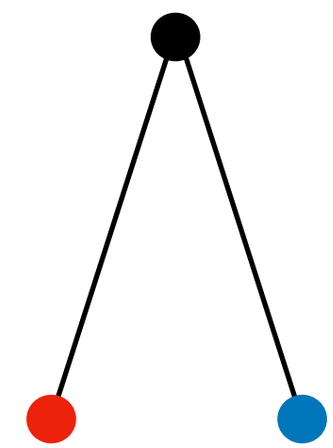
Li-and-Stephens HMM



- What are the observables and the hidden states?
- What are the transition probabilities and emission probabilities?

Modified Li-and-Stephens model





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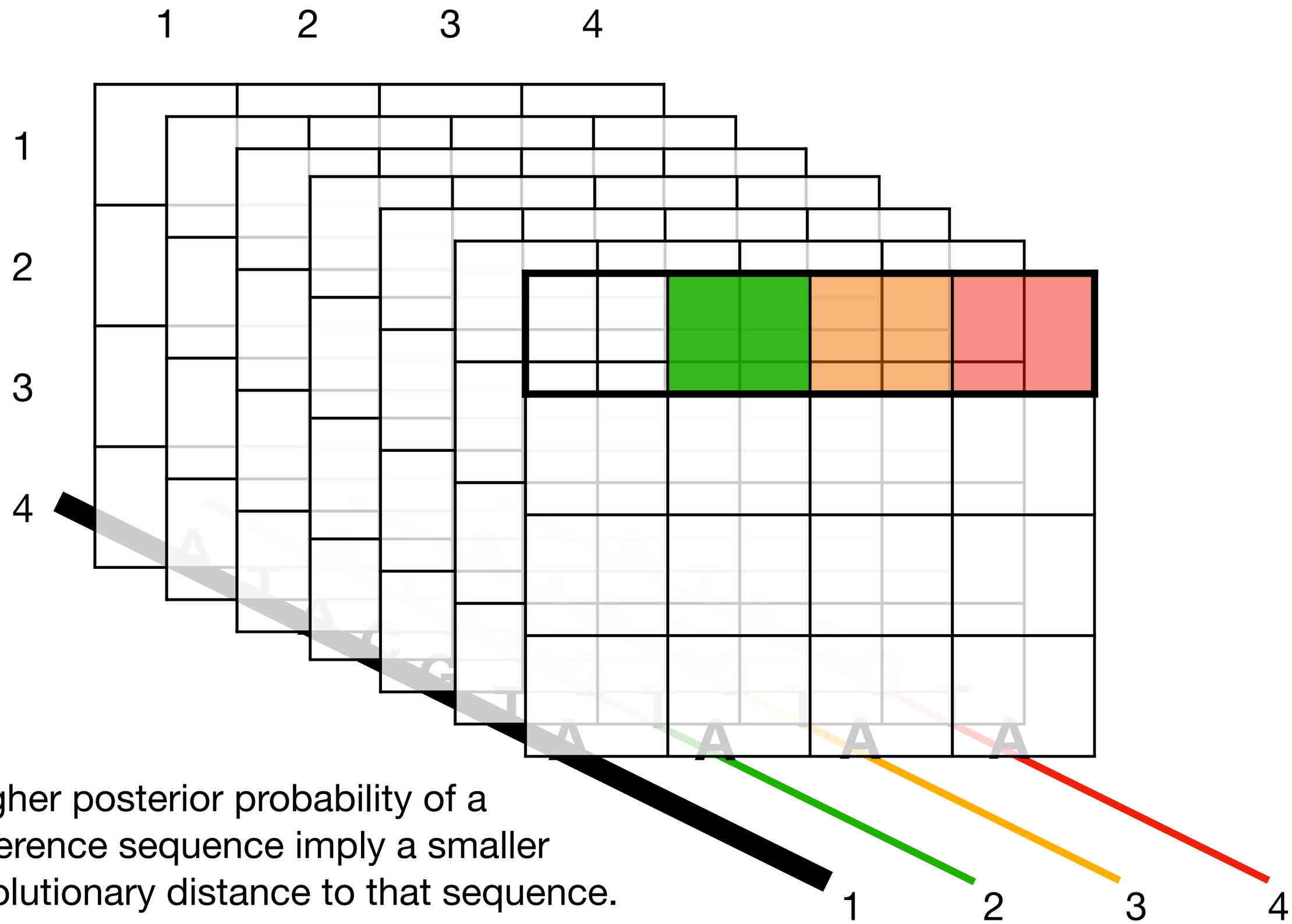
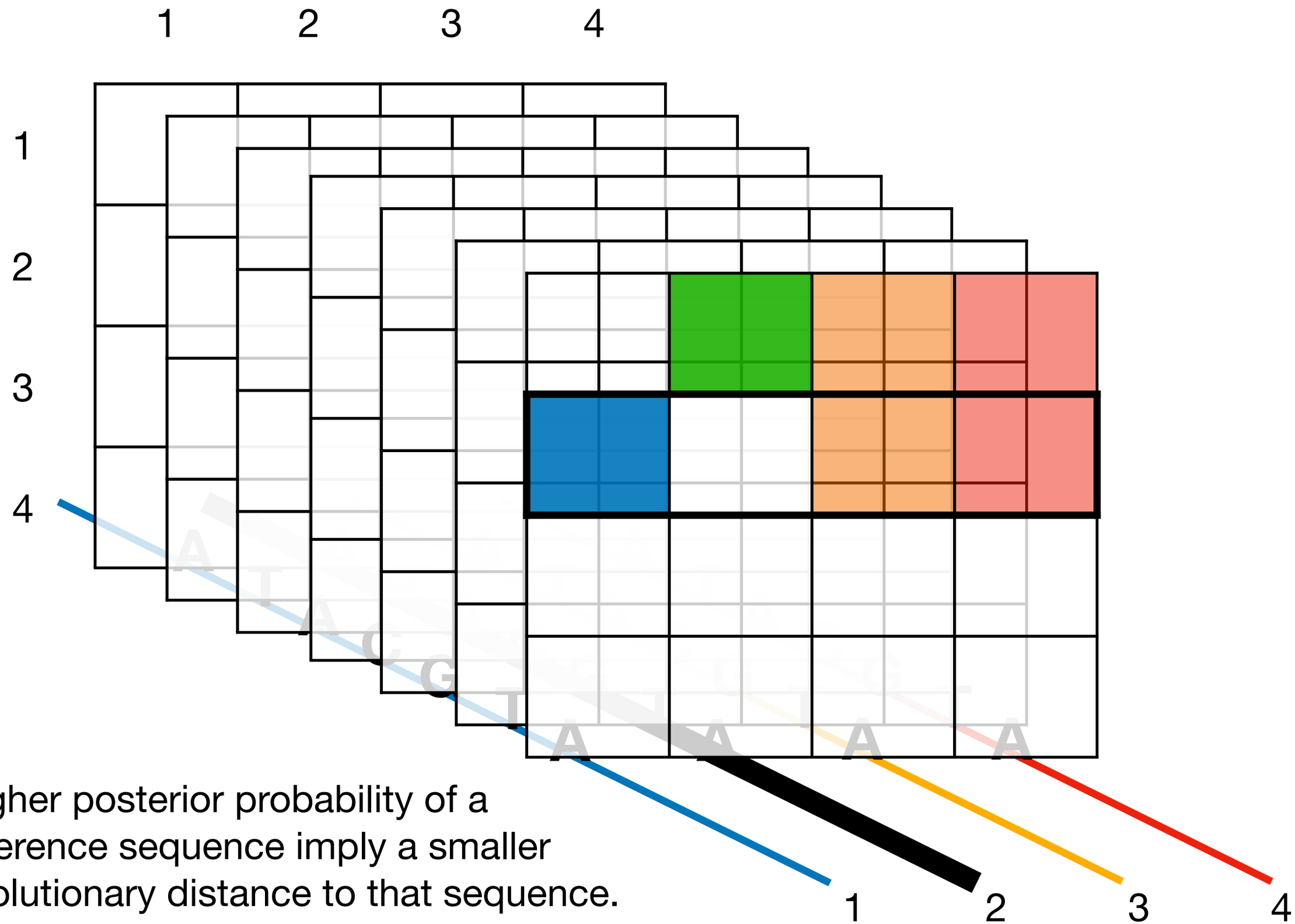
Sequence clustering



Higher posterior probability of a reference sequence imply a smaller evolutionary distance to that sequence.

Sequence clustering



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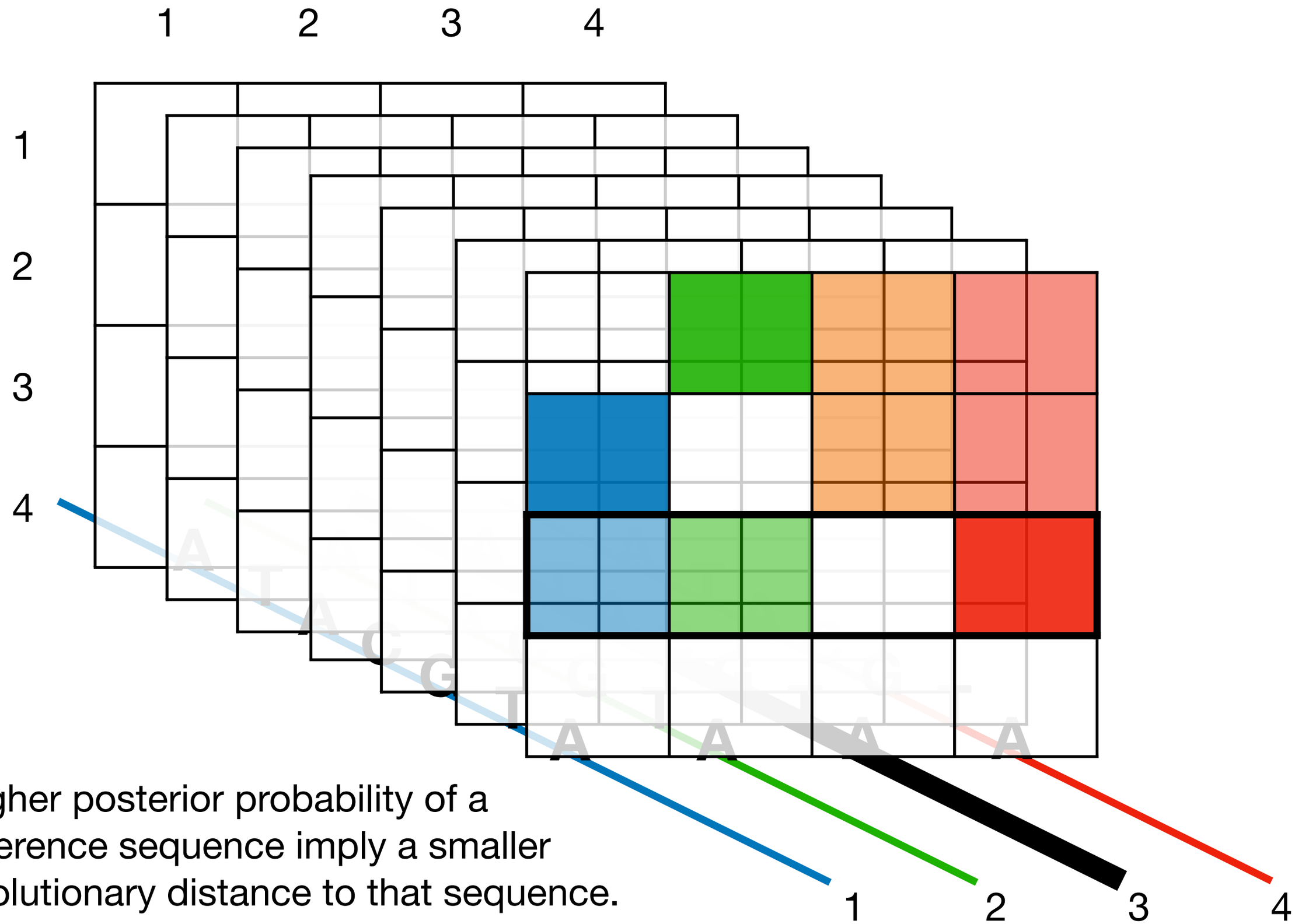

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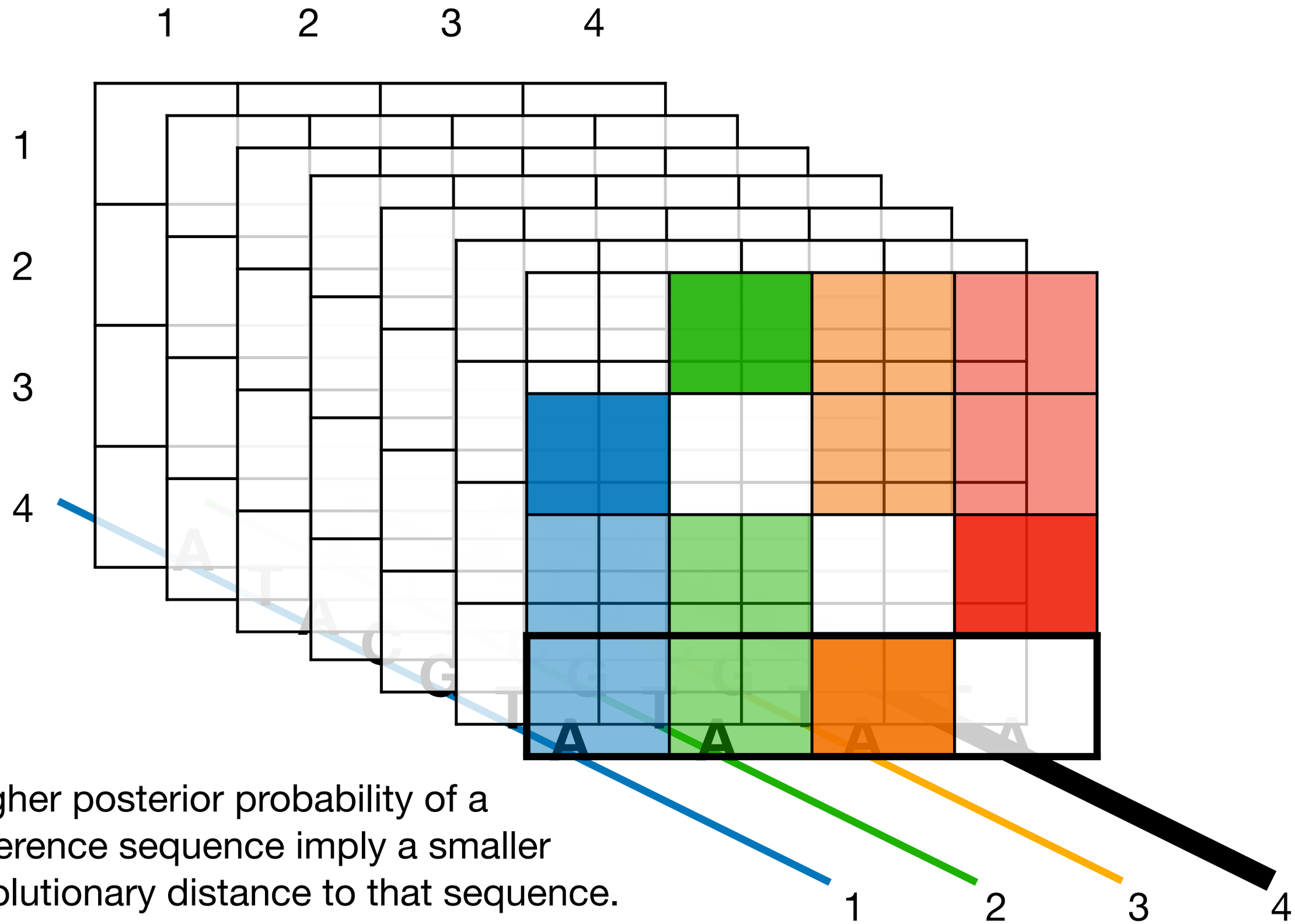

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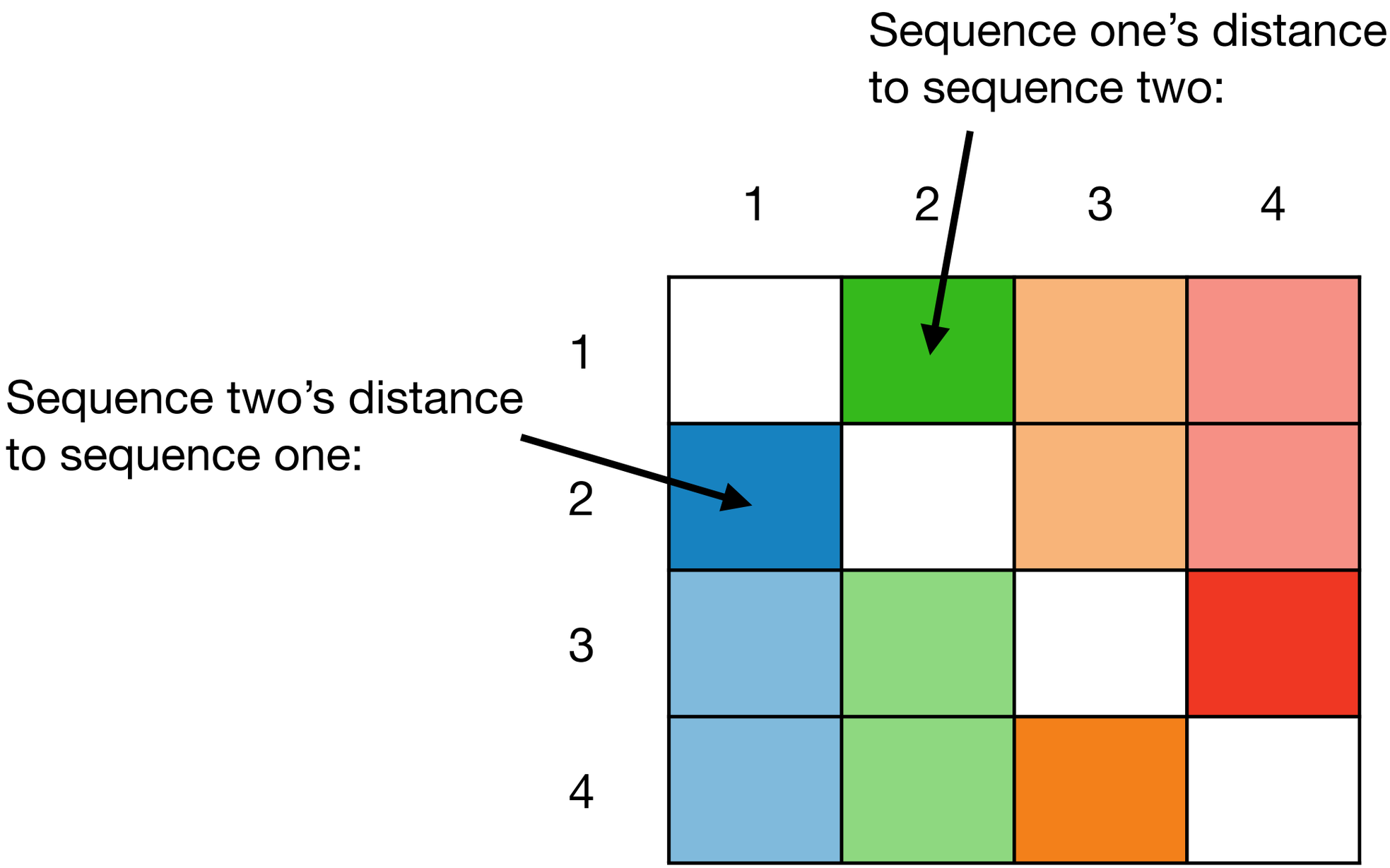


Higher posterior probability of a reference sequence imply a smaller evolutionary distance to that sequence.

Sequence clustering

Using an asymmetric matrix

Step 2

-	1	1
1	-	0
4	3	-

$$\arg\min_j d(\{1,2\},j) = 3,4$$

$$\arg\min_j d(3,j) = 4$$

$$\arg\min_j d(4,j) = 3$$

Step 1

-	2	2	2
0	-	0	0
1	1	-	0
4	4	3	-

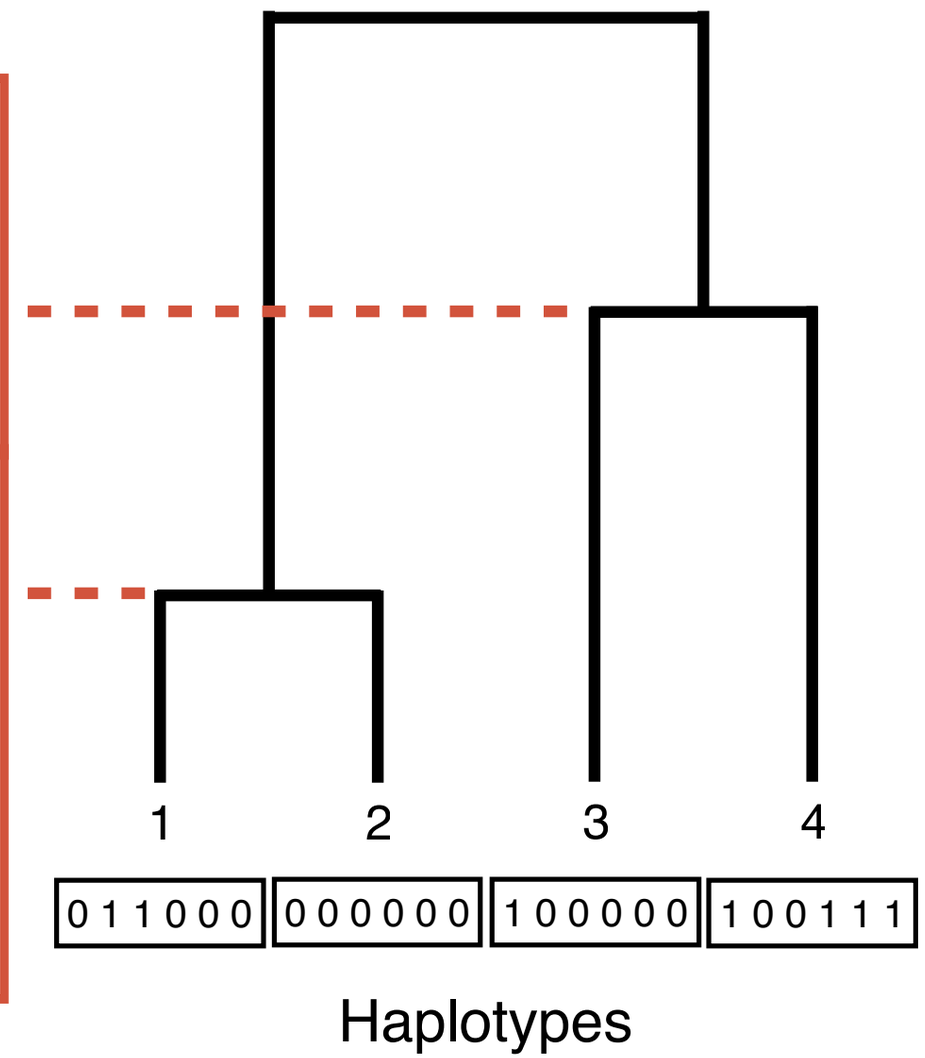
$$\| \; d$$

$$\arg\min_j d(1,j) = 2,3,4$$

$$\arg\min_j d(2,j) = 1,3,4$$

$$\arg\min_j d(3,j) = 4$$

$$\arg\min_j d(4,j) = 3$$



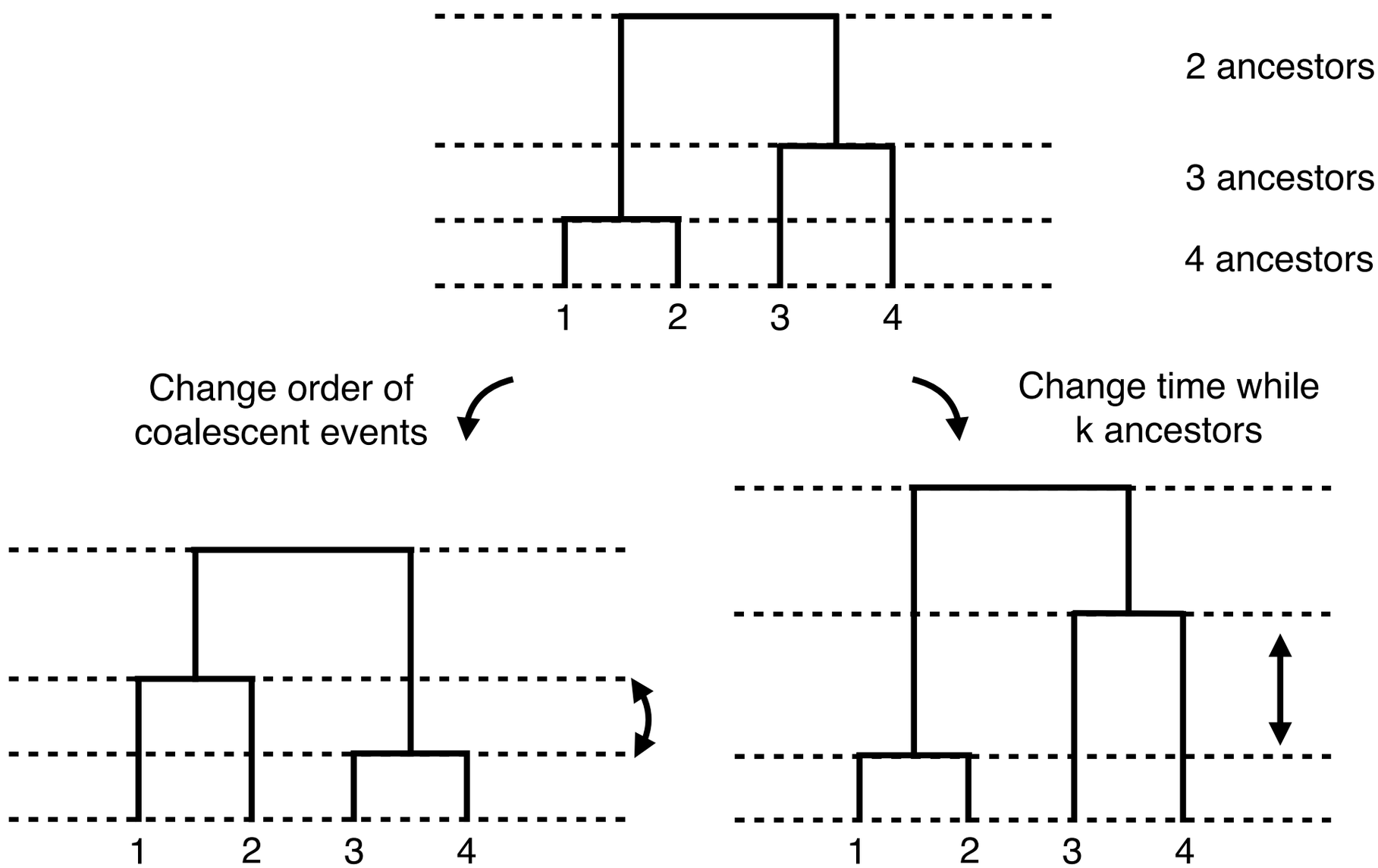
Making an initial tree sequence

Using clustering

1. Start from first base and use metric to construct tree.
2. Continue along alignment and keep the tree until we encounter a mutation that cannot be mapped onto the tree.
3. Make a new tree from the distance matrix at that position.
4. Keep going…

Estimate branch lengths

Using MCMC



Discussion topics

RELATE

1. How does RELATE compute the changing relationship of samples along the genomic alignment?

2. What is the Li-and-Stephens model and how does it work? Is it a HMM? What are the states, transitions and emissions?

3. How does RELATE build the trees along the genome?

4. How does RELATE find the branch lengths of the trees?

5. What are examples of the cool things you can do once you have all the trees?